

- ① Define distinguishable and indistinguishable state.
minimize the following DFA's

(i) δ	0	1	δ	a	b
$\rightarrow A$	B	E	$\rightarrow A$	B	F
B	C	F	B	G	C
* C	D	H	* C	A	C
D	E	H	D	C	G
E	F	I	E	H	F
* F	G	B	F	C	G
G	H	B	G	G	E
H	I	C	H	G	C
* I	A	E			

(i)

(ii)

- ② (i) State and prove Pumping lemma for a regular language.
(ii) Show that the following language are not regular:

- $L = \{ww^R \mid w \in (a+b)^*\}$
- $L = \{w \mid n_a(w) = n_b(w)\}$
- $L = \{a^n b^n \mid n \geq 0\}$
- $L = \{a^{n!} \mid n \geq 0\}$

- 3) Define context free grammar. Design a CFG for the

- $L = \{a^n b^{2n} \mid n \geq 1\}$
- $L = \{a^n b^n \mid n \geq 1\}$
- $L = \{ww^R \mid w \in \{a,b\}^*\}$
- $L = \{\text{set of all palindromes over } \{a,b\}\}$

- 4) Consider the grammar G with production, obtain most derivation (LMD), right most derivation (RMD), parse tree for some string for the following CFG

$$(a) S \rightarrow AbB$$

$$A \rightarrow Aa / \epsilon$$

$$B \rightarrow aB / bB / \epsilon$$

for the string "aaabab"

(i) generate grammar for the language $L = \{a^n b^{n+1} \mid n \geq 0\}$

$$L = \{b, abb, aabbb, aaa.bbbb, \dots\}$$

production:-

$$S \rightarrow abb$$

$$S \rightarrow b$$

$$S \rightarrow aSb$$

$$S \rightarrow aSb$$

$$S \rightarrow aSb$$

(ii) $L = \{w \mid |w| \bmod 3 = 0\}$ over $\Sigma = \{a, b\}$

$$L = \{a, aa, aaaa, aaaaa, \dots\}$$

$$S \rightarrow a$$

$$S \rightarrow aa$$

$$S \rightarrow Saaa$$

(iii) $L = \{a^n b^m c^k \mid m = n+k\}$ over $\Sigma = \{a, b, c\}$

$$L = \{abbc, aabbbbcc, \dots\}$$

$$a^n b^m c^k = a^n b^{n+k} c^k$$

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow bBc \mid bc$$

6) How to convert ambiguous grammar to unambiguous grammar.

7) Consider the grammar

$$E \rightarrow E+E | E-E | E * E | E/E | (E) | \epsilon$$

$$I \rightarrow 0 | 1 | a | b | Ia | Ib | Io | I1$$

and compute the string $(a1 + b0 * 00)$

(a) Is this ambiguous grammar?

(b) ~~is~~ convert to unambiguous grammar.

8) (b) $E \rightarrow E+E | E * E | E/E | \epsilon$

(a) Show that the given ambiguous grammar?

(b) convert the grammar to unambiguous grammar and construct parse tree for ambiguous grammar

9) Define PDA. obtain PDA to accept the language $L(M) = \{w \in \Sigma^* \mid w \in (a+b)^*\}$, where w^R is reverse of w by a final state.

(b) Convert the following CFG to PDA

$$S \rightarrow aABC$$

$$A \rightarrow aB | a$$

$$B \rightarrow bA | b$$

$$C \rightarrow a$$

10) (a) Explain the working of a PDA with a diagram.

(b) Design a PDA for accepting $a^n b^{2n}$

(c) Convert the following CFG to PDA.

$$E \rightarrow E+E | E * E | \epsilon$$

11) Discuss the Language accepted by a PDA. Design a PDA to accept the following language:

$L = \{0^{2n} 1^n \mid n \geq 1\}$. Draw the transition diagram for the constructed PDA. Also show the moves made

COMPASS PDA for the string "000111"

(b) $S \rightarrow aB \mid bA$

$A \rightarrow aS \mid bAA \mid a$

$B \rightarrow bS \mid aBB \mid b$

for the string $aaabbaabba$

(c) obtain the CFG for the RE $(a+ b)^+ ab (a+ b)^+$

(d) obtain the derivation and the parse tree for the string
 $w = id + id * id$

$$E \rightarrow E + E$$

$$E \rightarrow E * E$$

$$E \rightarrow id$$

(e) derive the string 011100 from the following grammar.

$$S \rightarrow 0S1S \mid 1S0S \mid \epsilon \text{ and draw the parse tree.}$$

Explain Ambiguous grammar? Show that the following grammar is ambiguous.

(i) $E \rightarrow E + E$

$$E \rightarrow E - E$$

$$E \rightarrow E * E$$

$$E \rightarrow E / E$$

$$E \rightarrow (E)$$

$$E \rightarrow id$$

$$w = id + id * id$$

(ii) Find LMD and RMD for the string $+ * - x y x y$ and write parse tree

$$E \rightarrow + E E$$

$$E \rightarrow * E E$$

$$E \rightarrow - E E$$

$$E \rightarrow x / y$$